

CELO **PROOF OF SHIP** · SEASON 8

Sentinel8004

The trust layer for Celo's ERC-8004 agent ecosystem. It scans every registered agent, scores it 0–100 on real signals, and writes verifiable attestations on-chain.

9,400 agents scored

3,541 attestations on mainnet

deterministic · no LLM

THE PROBLEM

Anyone can register an agent. Most register junk.

Celo's ERC-8004 IdentityRegistry shipped with **no quality or anti-Sybil layer**. The ReputationRegistry is already being gamed by sock-puppet feedback. `getSummary` returns attacker-inflated numbers on mainnet today.

SCALE

9,400 agents

registered on the IdentityRegistry, growing weekly, with zero trust signal attached.

MASS CLONING

6,934 agents, one owner

a single address mass-registered 6,934 agents to farm presence. Every one is capped at 15.

REPUTATION GAMING

1,797 sock puppets

fake wallets inflating three agents' on-chain reputation scores.

THE APPROACH

Five deterministic scoring layers, 0–100

L1 · 0–25

Registration

Metadata quality & completeness.

L2 · 0–25

Liveness

Endpoint probes: is it reachable?

L3 · 0–25

On-chain

Wallet behavior & transaction history.

L4 · 0–25

Sybil

Spam & clone detection across owners.

L5 · 0–15

Reputation

Existing on-chain feedback, Sybil-discounted.

Fully **deterministic**: no LLM, no randomness. Every score is reproducible and auditable, backed by an IPFS-pinned evidence report.

WHY IT CAN'T BE GAMED

Circuit breakers cap manipulators

- A **mass registrar** can't score above **15**, no matter how polished its metadata looks.
- Clone, dead-endpoint, and negative-reputation breakers each hard-cap the score.
- Weight changes barely move outcomes: rankings hold at **Spearman ρ 0.944–0.997** across 5 configs.

78.5%

of all 9,400 agents trip at least one circuit breaker.

 ≤ 15

hard ceiling for a detected mass registrar.

MEASURED ON MAINNET

The fraud Sentinel8004 surfaced

6,934

agents mass-registered by a single address.

1,797

sock-puppet wallets inflating three agents' reputation.

9,108

of 9,400 agents score below 30/100 (avg score 17.7).

Concrete example: **Toppa (#1870)** presents clean metadata and a 95-style profile, but 431 sock puppets boosted it. Sentinel scores it **80** and flags the manipulation.

ON-CHAIN PROOF

3,541 trust attestations, live on Celo mainnet

3,541

attestations written to the ReputationRegistry, each IPFS-backed.

#1853

Sentinel8004 is itself a registered ERC-8004 agent.

~0.009

CELO per attestation write (gas-estimated, batched).

REPUTATIONREGISTRY · CELO MAINNET

0x8004BAa17C55a88189AE136b182e5fdA19dE9b63

Every score is permanent, verifiable, and traceable to the exact evidence report that produced it.

INTEGRATION

Three ways to consume a trust score

FOR AGENTS

MCP server

check_agent_trust,
list_flagged_agents,
get_agent_report: query scores
programmatically.

FOR CONTRACTS

On-chain read

Read attestations straight from the
ReputationRegistry to gate interactions, no
extra infra.

FOR HUMANS

Live dashboard

Browse, search, and inspect every agent's
0–100 score and evidence at the public
site.

Any app, agent, or contract can gate who it interacts with, **without building its own trust evaluation.**

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What's new in June 2026

- **Full-registry rescan: 3,766 → 9,400 agents.** Scoring coverage now spans the entire IdentityRegistry.
- **~14× faster scorer**, concurrent + checkpointed, resumable mid-scan.
- **Honest dashboard refresh:** every surface now reports the same verified numbers (9,400 / 3,541 / 78.5%).
- **Re-validated calibration** on the full set: rankings stable at p 0.944–0.997 across weight configs.

ROADMAP

Next milestones

M2

Attest the new set

Write ReputationRegistry attestations for the ~5,600 newly-scanned agents in gas-estimated batches (test tx first).

M3

Public read API

REST endpoints (GET /agent/:id, /flagged) so any app queries a score over HTTP, no local MCP server.

INFRA ASK

Whitelisted scorers

A standard to identify legitimate scorers so registry reads aren't trivially Sybil-able.

Sentinel8004 measures the exact Sybil problem proof-of-humanity primitives are built to solve. It is the missing quality layer for the agent economy.

SENTINEL8004 · THE TRUST LAYER FOR CELO AGENTS

Scanned 9,400 agents.
Wrote 3,541 scores on-chain.
Caught the fraud nobody else flagged.

DASHBOARD

yonkoo11.github.io/sentinel8004

CODE

github.com/Yonkoo11/sentinel8004